

Water hammer response characteristics of wellbore-fracture system: Multi-dimensional analysis in time, frequency and quefreny domain

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ARTICLE INFO

Keywords:

Hydraulic fracturing
Water hammer
Fracture diagnostics
Cepstrum analysis
Fourier transform

ABSTRACT

Water hammer fracture diagnostics is a promising fracture diagnostic method, which is nonintrusive, economical, timely performed and easily operated. In present studies, the propagation behavior of the water hammer in the wellbore-fracture system remains unclear, which brings uncertainty to locating the hydraulic-stimulated fractures. In this paper, the transient flow model was used to describe the water hammer pressure fluctuation, which was generated after the pump shut down during the hydraulic fracturing treatment, and the wellbore-fracture system was built by introducing the branch pipe to the wellbore as the fracture. The method of characteristics (MOC) was used to solve this model, then was validated by the computational fluid dynamics software FLUENT. Combining water hammer signal analysis in time, frequency, and quefreny domain, the fracture effect on the water hammer response characteristics in the wellbore-fracture system was investigated in multiple dimensions. We then built eight cases to simulate the water hammer pressure fluctuation in different wellbore-fracture systems, including the existence, the varied location and the varied length of the fracture and conducted the multi-dimensional analysis. After that, results were discussed and the response characteristics of the water hammer in the wellbore-fracture system were summarized. From this work, a multi-dimensional analysis method in time, frequency and quefreny domain is proposed. Meanwhile, the water hammer response characteristics of the wellbore-fracture system are sufficiently presented for the first time to accurately investigate the water hammer pressure propagation in the wellbore-fracture system and provide useful insights for water hammer diagnostics in field application.

Credit author statement

Yang Qiu: Conceptualization, Methodology, Data analysis, Visualization and Writing. **Xiaodong Hu:** Funding acquisition, Supervision, Conceptualization, Writing-Review and Editing. **Fujian Zhou:** Supervision. **Zhuolong Li:** Validation. **Yujiao Li:** Software. **Yinghao Luo:** Data Curation.

1. Introduction

Fracture diagnostics is an important and well-concerned research topic nowadays in the development of unconventional reservoir since the wide application of hydraulic fracturing treatment (Parkhonyuk et al., 2018a; Liu et al., 2021). Hydraulic fracturing treatment is crucial to the successful exploitation of unconventional oil and gas resources (William et al., 2018; Dong et al., 2019; Sadykov et al., 2019; Guo et al.,

2020; He et al., 2020; Wang et al., 2020). A large amount of fracturing fluid and proppants are pumped underground at the surface. Fractures are generated under the high pressure of the fluid, which provide a conductive pathway for oil and gas transportation. The scale of the fracture network after hydraulic fracturing treatment directly determines the final production (Ciezobka et al., 2016; Iriarte et al., 2017; Dung et al., 2021). As reported, there are many perforation clusters within horizontal well completion, which do not contribute to oil and gas production, and poor stimulation seriously affects the final production (Barraza et al., 2017; Parkhonyuk et al., 2018a). Methods are required to evaluate the effectiveness of the hydraulic fracturing and to provide real-time guidance and optimizations to completion strategies. Fracture diagnostics is an effective evaluation method.

Water hammer fracture diagnostics has been developed and validated in the field application since 2016 (Bogdan et al., 2016; Korkin et al., 2020). Compared with traditional intrusive diagnostics, the water

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<https://doi.org/10.1016/j.petrol.2022.110425>

Received 6 September 2021; Received in revised form 24 January 2022; Accepted 16 March 2022

Available online 23 March 2022

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hammer fracture diagnostics is easy to operate. Meanwhile, the economic benefit of this technique is high, providing decisions optimizing fracturing stimulation in time and satisfying the requirements of the large-scale application in field. The process of the water hammer fracture diagnostics is as follows. After the pumps shut down at the surface, water hammer wave is generated. Propagating along the wellbore, water hammer pressure wave will interact with downhole events (such as hydraulic-connected fractures or case leaking) and reflect (Liang et al., 2017; Parkhonyuk et al., 2018b). These events, altering the water hammer response characteristics, can be located by conducting cepstrum analysis of the water hammer pressure data collected by pressure gauge at the surface. The cepstrum of pressure signal includes the reflection time of events and can be used for echoes detection (Parkhonyuk et al., 2018b). Borrowing from principles of the convolution model of recorded seismogram, reflection time of downhole events like fractures or case leaking can be determined by conducting cepstrum analysis. Therefore, cepstrum analysis is the key method for water hammer diagnostics.

The mechanism of the water hammer propagation is complicated (Hou et al., 2021). The water hammer responds differently while the fracture is introduced to the wellbore system. Understanding the water hammer response characteristics in the wellbore-fracture system is a challenge. However, water hammer propagation behavior has not been fully understood, especially in the wellbore-fracture system after hydraulic fracturing treatment. This brings uncertainty to cepstrum analysis of field water hammer data. In addition, noises exist in water hammer data, which bring difficulty in the following cepstrum analysis. How to understand the water hammer response characteristics accurately in a wellbore-fracture system is a challenging issue for the analysis of water hammer data cepstrum.

Many field studies applying the water hammer diagnostics, collecting pressure data by high-frequency transducers at the wellhead and locating the position of the stimulated fractures, have been reported in the literature. The fundamental frequency and harmonic series of water hammer signatures were used to calculate the hydraulic length of the wellbore, which indicated the location of hydraulic fractures (Clark et al., 2018). The water hammer pressure data were used to obtain the period, identifying the wellbore boundary conditions, including treatment stage isolation, casing failure depth or extended length of the fracture (Dung et al., 2021). Since 2016, a new method called high-frequency pressure monitoring (HFFM), based on the signal processing technique, analyzing the wellbore pressure oscillations recorded at the wellhead has been developed and applied in hydraulic fracturing monitoring (Bogdan et al., 2016; Klem et al., 2017; Rutzinger et al., 2017; Kabannik et al., 2018; Parkhonyuk et al., 2018a, 2018b, 2018c; Panjaitan et al., 2018; Korkin et al., 2020). Combining the reflection time and the wave speed, the depth of downhole events, such as ball landing, location of the zone accepting the fluid, diverter efficiency and the casing failure depth, are calculated and determined. This method monitors the fracturing treatment in real time and makes it possible to conduct cost-effective strategies in time during the treatment.

However, the real features of the fracture after fracturing remains unknown underground. The corresponding response characteristics of the water hammer are not clear. Moreover, no study has been performed so far to investigate the water hammer propagation behavior in the wellbore-fracture system. Only the water hammer propagating in the wellbore system is considered to simplify the analysis. As the fracture is introduced to the wellbore system, the effects that the pressure wave propagating in the fracture in the wellbore-fracture system are not taken into consideration. Thus, experimental and numerical studies are needed to explain the water hammer response characteristics in a wellbore-fracture system.

Many experimental and numerical studies of water hammer have also been reported. Water hammer propagation behavior in pipeline system was experimentally studied and numerically simulated (Nikpour et al., 2014; Wang et al., 2015; Mounir and Ali, 2019; Rahul and Arun,

2020). This pressure behavior was very important in long-distance transportation and valve operation. Thus, many experimental and numerical studies were conducted to prevent mistake operation in valve switching and to ensure the pipeline and pumps safety (Asadiniazzi et al., 2010; Pierre and Gudmundsson, 2011; Wu et al., 2015). The pressure transient behavior was also studied in pipeline blockage and leakage. According to different pressure transient responses, algorithms were developed and applied to block or leak detection and evaluate the blockage or leakage characteristics (Qiang et al., 2018; Lang and Yuan, 2020; Zhang et al., 2021; Fu et al., 2021). All the studies can help to understand the water hammer propagation behavior.

However, all these researches are studied in the pipeline system. No branched pipe exists in this system. The water hammer response characteristics of the wellbore-fracture system are completely different from those in the pipeline system. In addition, analysis of water hammer signatures mentioned above is only considered in time or frequency dimension. The water hammer propagation behavior in a complex system is complicated. Combining methods in multi-dimensions can help to better understand the water hammer response in a complex system.

In this work, we proposed a multi-dimensional analysis method in time, frequency and quefrequency domain for water hammer analysis. Meanwhile, the water hammer response characteristics of the wellbore-fracture system are sufficiently presented for the first time to accurately investigate the water hammer propagation in the wellbore-fracture system. First, the wellbore-fracture system was built by modifying the fracture as the branch pipe to the wellbore, and the water hammer propagation in the wellbore-fracture system was derived from the transient flow model. This model was then validated by the computational fluid dynamics software FLUENT. Second, eight cases were designed to simulate the water hammer fluctuation in different wellbore-fracture systems, including the existence of the fracture, the varied location of the fracture, and the varied length of the fractures in the system and conducted multi-dimensional analysis in time, frequency and quefrequency domain, respectively. Third, the results were discussed and the water hammer response characteristics in the wellbore-fracture system were summarized. This study can provide clear insights into the water hammer fracture diagnostics and improve the identification of fractures and other downhole events.

2. Methodology

First, the mathematical model was presented including the transient flow model simulating water hammer fluctuation in the wellbore-fracture system and the multi-dimensional analysis method. Second, the configuration of the physical model was presented. Third, the model was validated with FLUENT.

2.1. Transient flow model

2.1.1. Method of characteristics

The pressure transient behavior like water hammer in the wellbore is governed by the motion and continuity equations. A more detailed review can be found in Carey et al. (2015). By describing pressure and velocity, P and V , in terms of pressure head and discharge, H and Q , and the governing equations can be written as:

$$\frac{\partial H}{\partial t} + \frac{a^2}{gA} \frac{\partial Q}{\partial x} = 0 \quad (1)$$

$$\frac{\partial Q}{\partial t} + gA \frac{\partial H}{\partial x} + \frac{fQ|Q|}{2DA} = 0, \quad (2)$$

where a is the wave speed (m/s), A is the area of pipe cross-section (m²), H is the pressure head (m), Q is the discharge (m³/s), D is the pipe diameter (m), and f is the friction coefficient.

The pressure wave speed in the pipe can be written as:

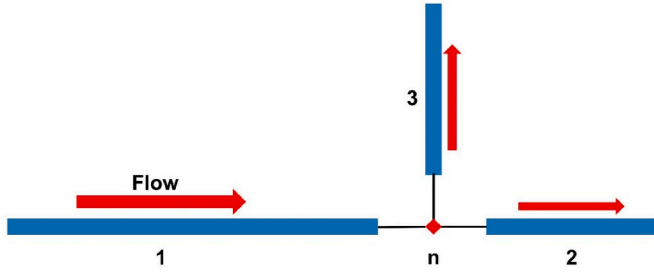


Fig. 1. The boundary condition of the intersection point between the wellbore and the fracture.

$$a = \sqrt{\frac{K/\rho}{1 + \left(\frac{K}{E}\right)\psi}} \quad (3)$$

where K is the fluid bulk modulus, ρ is fluid density (kg/m^3), E is pipe Young's modulus of elasticity and ψ was determined from the expression for a thin-walled elastic pipe anchored against the longitudinal movement,

$$\psi = \frac{D}{e} (1 - \nu^2), \quad (4)$$

where e is the thickness (m) and ν is the pipe Poisson's ratio. However, to simplify the analysis, the wave speed in the wellbore was specially set in this study. More details can be seen in the following part model configuration.

The water hammer equations Eq. (1) and (2) are a pair of quasi-linear hyperbolic partial differential equations, which can't be solved directly. The method of characteristics (MOC) is used to transform the two partial differential equations into four ordinary differential equations along the characteristic lines C^+ and C^- in the x - t plane, as shown in Eq. (5) and Eq. (6):

$$C^+ \begin{cases} \frac{dQ}{dt} + \frac{gA}{a} \frac{dH}{dx} + \frac{fQ|Q|}{2DA} = 0 \\ \frac{dx}{dt} = a \end{cases}, \quad (5)$$

$$C^- \begin{cases} \frac{dQ}{dt} - \frac{gA}{a} \frac{dH}{dx} + \frac{fQ|Q|}{2DA} = 0 \\ \frac{dx}{dt} = -a \end{cases}. \quad (6)$$

The solutions of ordinary differential equations are not always valid in the x - t plane. Assuming that a is constant, they are only valid along the lines with the slope of $1/a$ and $-1/a$ in the x - t plane instead.

2.1.2. Finite difference method

The transient flow model can be solved numerically by the finite difference method. Dividing the wellbore into n sections with the length Δx , the time step can be determined by $\Delta t = \Delta x/a$ to ensure the stability of the model. Eq. (7) and Eq. (8):

$$C^+ : H_P = H_A - B(Q_P - Q_A) - RQ_P|Q_A|, \quad (7)$$

$$C^- : H_P = H_B + B(Q_P - Q_B) + RQ_P|Q_B|, \quad (8)$$

including second-order approximation, are the finite difference equations of Eq. (5) and Eq. (6) for C^+ and C^- , respectively.

B and R represent pipe characteristic impedance and pipe resistance coefficient, respectively.

$$B = \frac{a}{gA}, \quad (9)$$

$$R = \frac{f\Delta x}{2gDA^2}, \quad (10)$$

where f is the friction coefficient and Δx is the space step.

This model starts at $t = 0$ s. H and Q of each grid in the wellbore are known under initial steady-state conditions. Then H and Q at every Δt can be calculated from the known values at previous time steps $i-1$ and $i+1$. Therefore, the finite difference equation can be written as Eq. (11) to Eq. (12) according to time,

$$C^+ : H_i^{t+\Delta t} = C_P - B_P Q_i^{t+\Delta t}, \quad (11)$$

$$C^- : H_i^{t+\Delta t} = C_M + B_M Q_i^{t+\Delta t}, \quad (12)$$

where i refers to every intersection point. C_P , B_P , C_M , and B_M are constants, which can be calculated from Eq. (13) to Eq. (16):

$$C^+ : H_i^{t+\Delta t} = C_P - B_P Q_i^{t+\Delta t}, \quad (13)$$

$$C^- : H_i^{t+\Delta t} = C_M + B_M Q_i^{t+\Delta t}, \quad (14)$$

$$C^+ : H_i^{t+\Delta t} = C_P - B_P Q_i^{t+\Delta t}, \quad (15)$$

$$C^- : H_i^{t+\Delta t} = C_M + B_M Q_i^{t+\Delta t}. \quad (16)$$

Solving Eq. (11) and Eq. (12), yields:

$$H_i^{t+\Delta t} = \frac{C_P B_M + C_M B_P}{B_M + B_P}, \quad (17)$$

$$Q_i^{t+\Delta t} = \frac{C_P - C_M}{B_M + B_P}. \quad (18)$$

Eq. (17) and (18) are only valid within the pipe section that both C^+ and C^- equations are present. Additional boundary conditions are also required, containing the inlet on the surface, outlet at the end of the fracture, the downhole endpoint, or the pipe junction.

2.1.3. Boundary conditions

The main boundary conditions used in this simulation in this paper were inlet boundary, bottom hole zero-flow boundary, outlet pressure boundary, and the junction point between the wellbore and the hydraulic-connected fracture.

The pump is assumed to be the linear shutdown. Thus, the flow rate is decreased in a short time from the initial flow to zero. Then, the pressure head H at the inlet of each time step is solved by combining the C^- equation with the boundary condition of the linear pump shutdown as,

$$Q = \begin{cases} Q_0 \left(1 - \frac{t}{T_{\text{shutdown}}}\right), & t \leq T_{\text{shutdown}} \\ 0, & t > T_{\text{shutdown}} \end{cases}, \quad (19)$$

where Q_0 is the initial flow rate (m^3/s), T_{shutdown} is the total time the flow rate goes to zero (s), and Q (m^3/s) is the flow rate at time t (s).

Considering the outlet pressure boundary is a constant, the characteristic equation C^+ is used. Integrating with the outlet boundary condition, the outlet discharge Q is solved at each time step.

As shown in Fig. 1, the pressure head H at node n in pipe 1 (which is also the first point in pipe 2 and pipe 3) is equal, as shown in Eq. (20). Assuming that the fluid flow in from pipe 1 and out from pipe 2 and pipe 3, the total discharge Q should satisfy Eq. (21). The first subscript represents the label of the corresponding pipeline, while the second subscript represents the node number in each pipe. The superscript represents the computation time.

$$H_{1,n}^t = H_{2,1}^t = H_{3,1}^t = H^t, \quad (20)$$

$$Q_{1,n}^t - Q_{2,1}^t - Q_{3,1}^t = 0. \quad (21)$$

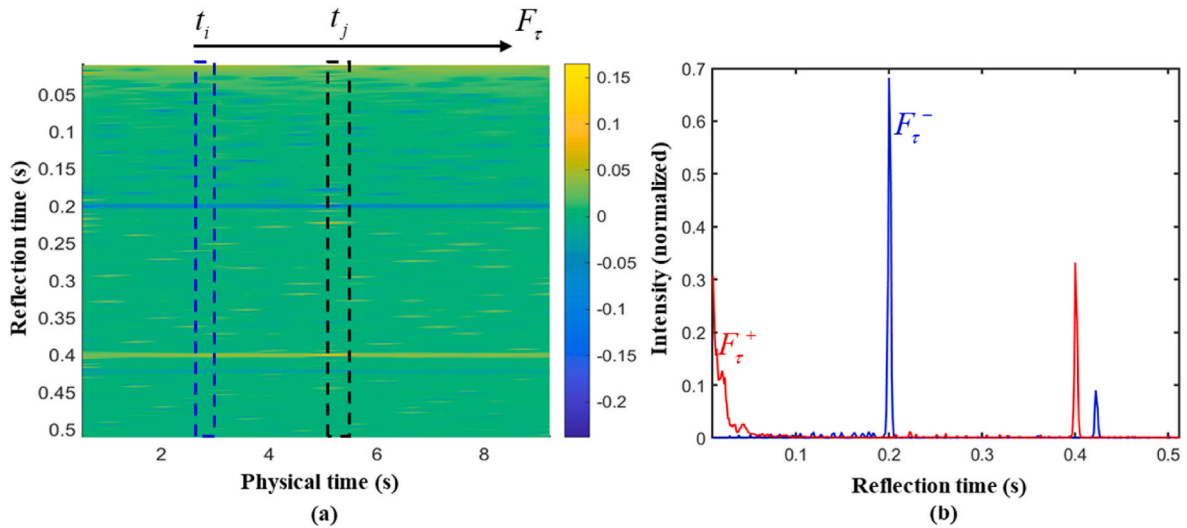


Fig. 2. The calculation process of the function of reflection time: (a) The cepstragram of water hammer signal; (b) The functions of reflection time.

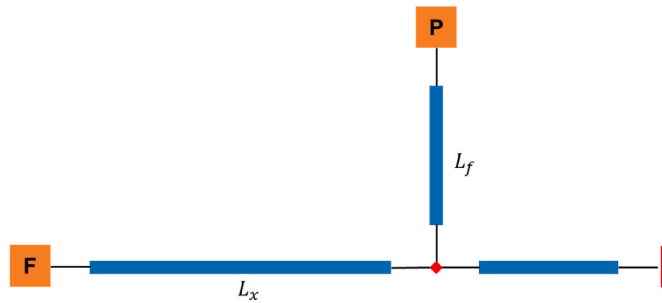


Fig. 3. The sketch of the wellbore-fracture system.

Table 1
The basic parameters for model validation.

Parameters	Fluent	MOC
Injection rate (m ³ /min)	0.06	0.06
Shut-in time (s)	0.005	0.005
Wellbore length, L_w (m)	10	10
Fracture location, L_x (m)	7	7
Fracture length, L_f (m)	5	5
Wave speed (m/s)	-	1490
Mesh size (m)	0.01	-
Mesh numbers	15,224	-
Wellbore diameter (m)	0.1	0.1
Pressure boundary (MPa)	0	0

During the hydraulic fracturing treatment, the fluid stopped at the bottom of the wellbore theoretically, for the presence of the well bottom or bridge. Integrating the zero-flow discharge with the C^+ equation, the pressure head H at the bottom can be calculated.

2.2. Multi-dimensional analysis model

In previous research, water hammer characteristics including period, amplitude, and decay rate were the important content of water hammer analysis. In this study, the key interest is the propagation behavior of the water hammer wave in the wellbore-fracture system. After exited at the wellhead, the water hammer wave propagates along the wellbore and reflects when the wave incidents on the hydraulic stimulated fracture or other downhole events. This phenomenon is the key point in our study. To understand the propagation behavior and reveal the relevant reflection events, analysis in time, frequency, and quefrequency domain was

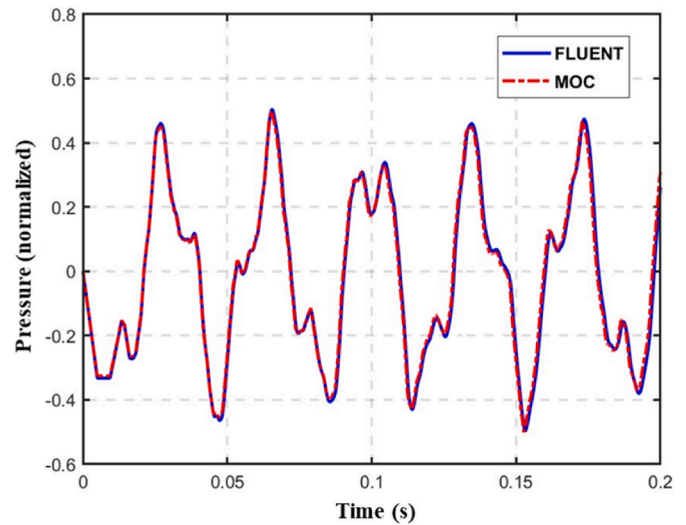


Fig. 4. Comparison of Fluent and MOC simulation results.

Table 2
Physical parameters and boundary conditions used in the simulation.

Case	1	2	3	4	5	6	7	8
Injection rate (m ³ /min)	6	6	6	6	6	6	6	6
Shut-in time (s)	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
Wellbore length, L_w (m)	150	150	150	150	150	150	150	150
Fracture location, L_x (m)	100	100	100	100	100	60	80	120
Fracture length, L_f (m)	0	25	50	100	200	100	100	100
Wellbore diameter (m)	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Pressure boundary (MPa)	10	10	10	10	10	10	10	10

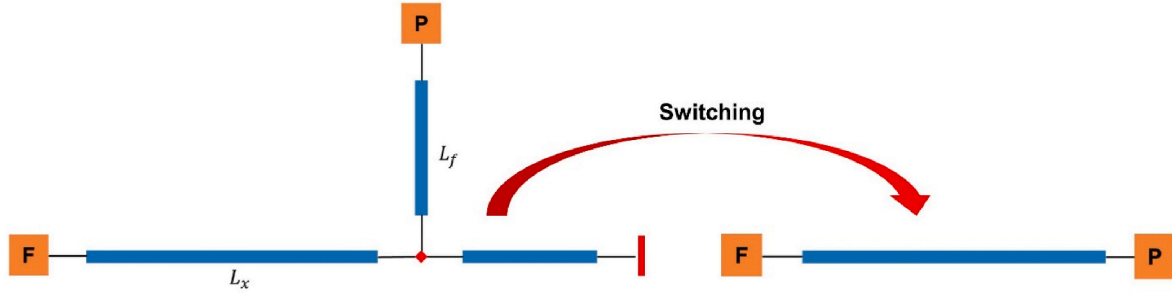


Fig. 5. The transformation of zero-length fracture in wellbore-fracture system.

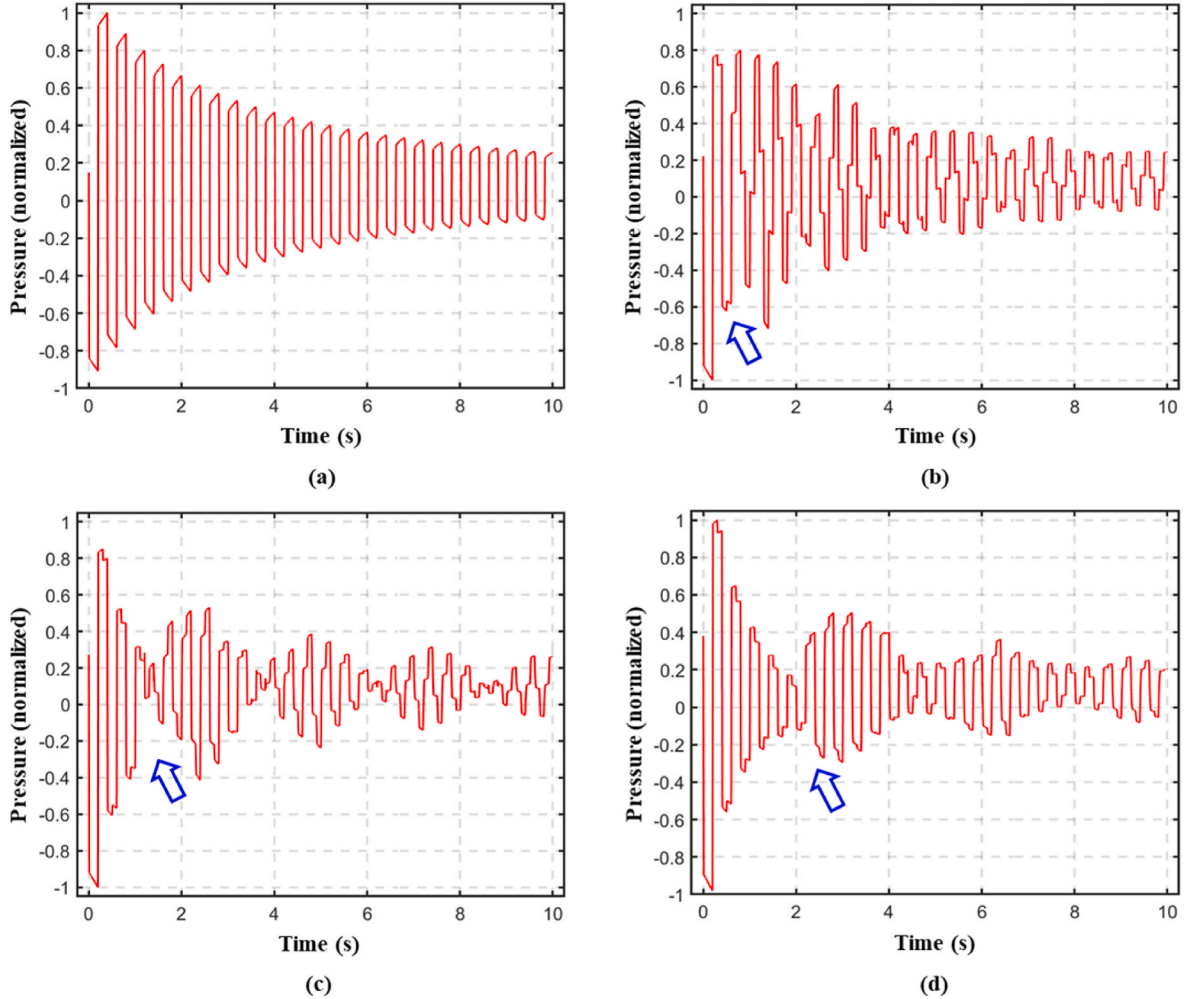


Fig. 6. The results of case 1 to 4 in the time domain: (a) case 1 fracture length: 0 m, location: 100 m; (b) case 2 fracture length: 25 m, location: 100 m; (c) case 3 fracture length: 50 m, location: 100 m; (d) case 4 fracture length: 100 m, location: 100 m.

conducted to the simulated water hammer fluctuation in the wellbore-fracture system by the fluid transient model mentioned above.

2.2.1. Time analysis

Water hammer signal, whether acquired from the wellhead transducers or collected from the simulation results, is the variable changing over time. The water hammer wave oscillates in the pipeline with period T and decays with time. However, the period identification would be difficult when the fracture is introduced to the wellbore system. In this study, the change points of the pressure trend were read, named as pressure response time. Every pressure response time at the wellhead corresponds to the water hammer reflection behavior. The reflection

location is calculated by

$$L_1 = \frac{a \times t_r}{2}, \quad (22)$$

where L_1 is the location of the reflection event (m), a is the wave speed in the pipeline and t_r is the time when the pressure trend is changed (s).

2.2.2. Frequency analysis

Water hammer is the free oscillation that is excited by the sudden change of the discharge, representing the resonance of the vibration system in the frequency domain. The complicated the vibration system

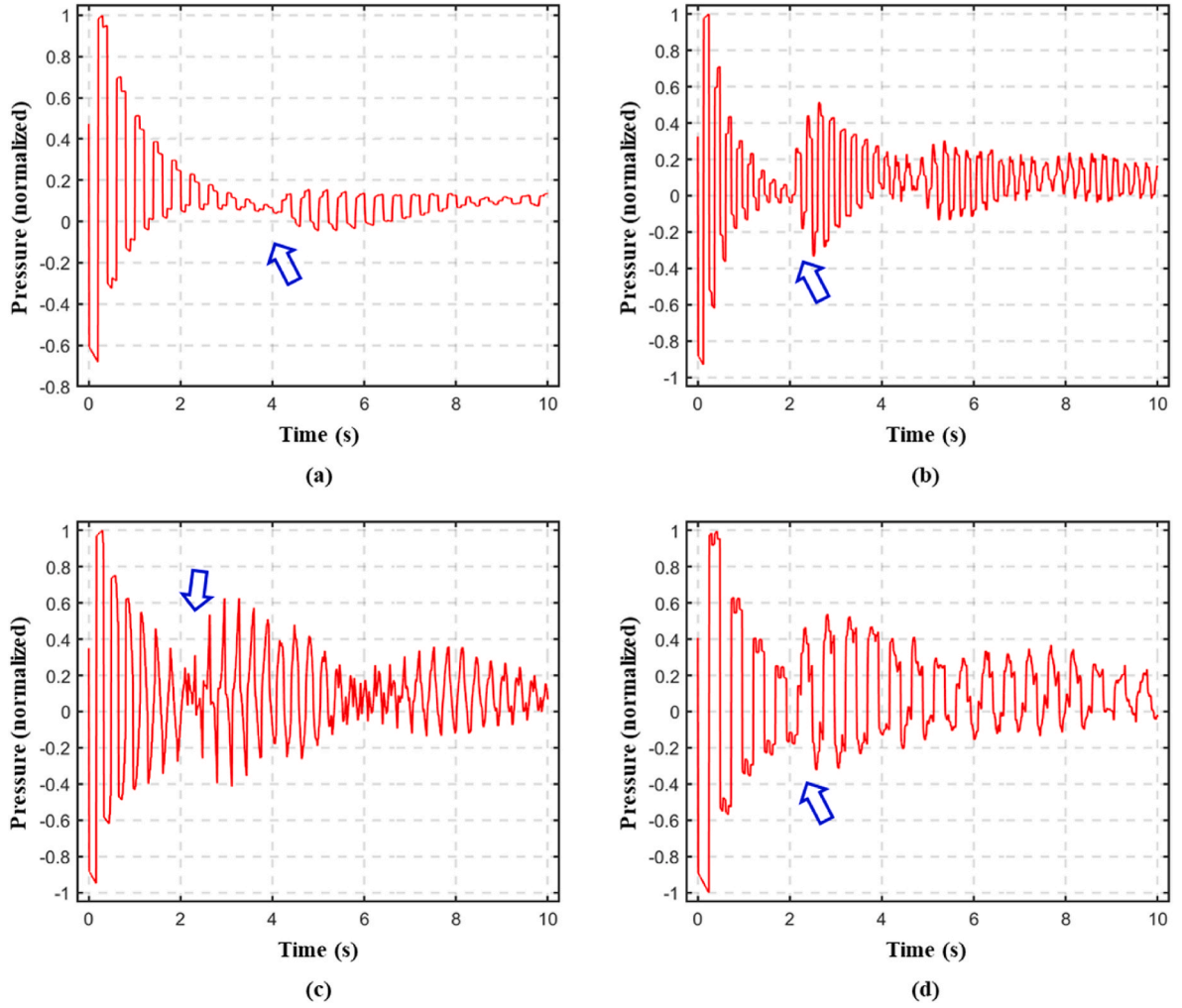


Fig. 7. The results of case 5 to 8 in the time domain: (a) case 5 fracture length: 200 m, location:100 m; (b) case 6 fracture length: 100 m, location:60 m; (c) case 7 fracture length: 100 m, location: 80 m; (d) case 8 fracture length: 100 m, location:120 m.

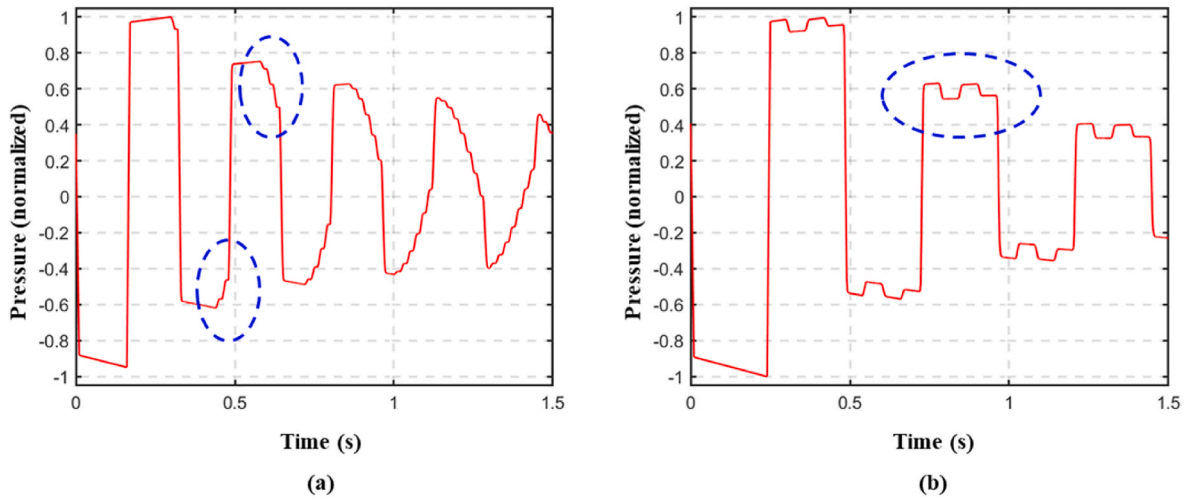


Fig. 8. The first 1.5 s detail results of case 7 and 8 in the time domain: (a) case 7 fracture length: 100 m, location: 80 m; (b) case 8 fracture length: 100 m, location:120 m.

was, the harder the water hammer period identification would be. Thus, another method is needed. Fourier transform is a mathematical tool to analyze the frequency characteristics of periodic or aperiodic signals

(Martinez et al., 2014). It can transform the signal from the time domain to the frequency domain for analysis, and it is solved by

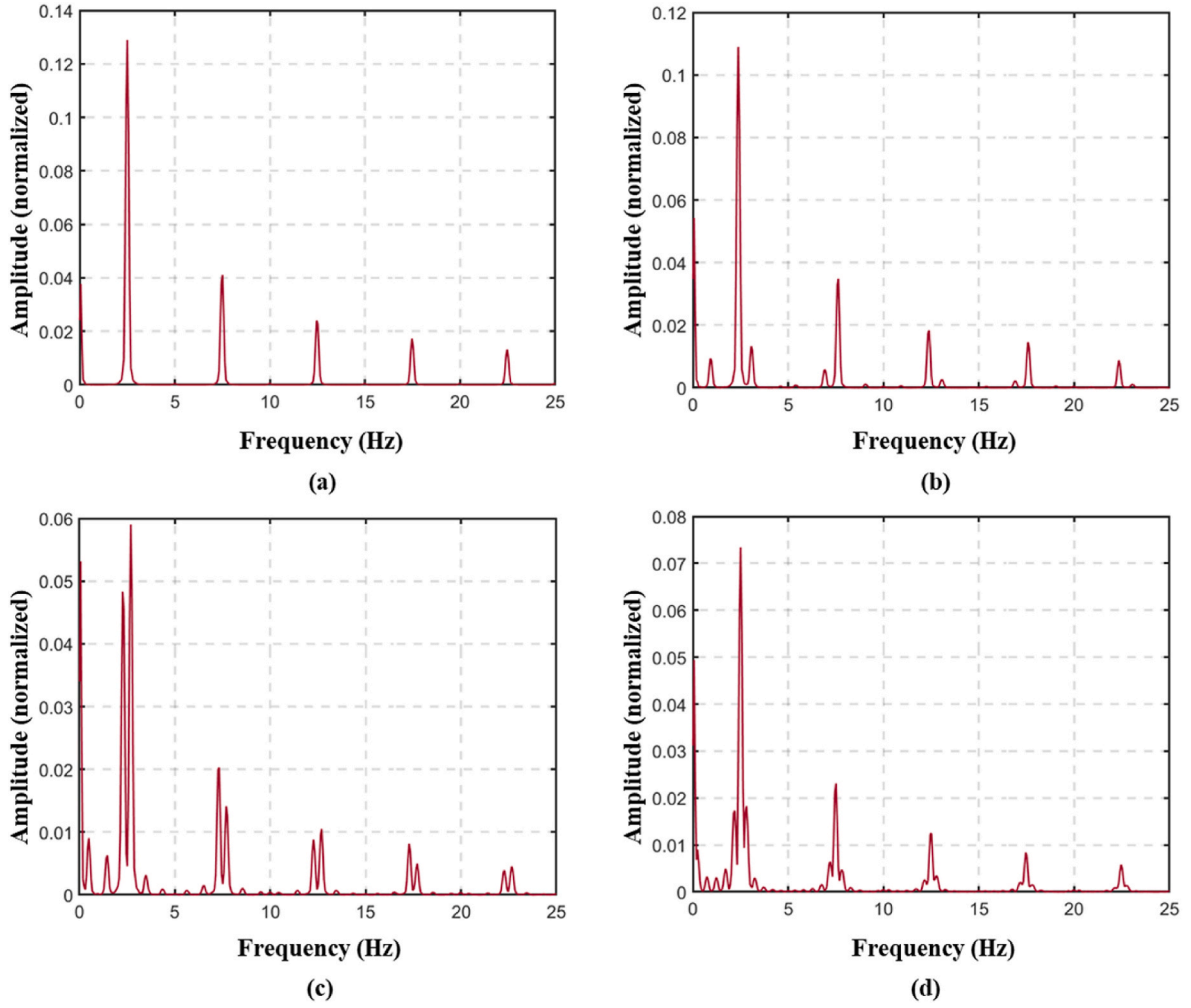


Fig. 9. The results of case 1 to 4 in the frequency domain: (a) case 1 fracture length: 0 m, location: 100 m; (b) case 2 fracture length: 25 m, location: 100 m; (c) case 3 fracture length: 50 m, location: 100 m; (d) case 4 fracture length: 100 m, location: 100 m.

$$\hat{x}(\omega) = \frac{1}{2\pi} \int x(t) e^{-i\omega t} dt, \quad (23)$$

where $x(t)$ and $\hat{x}(\omega)$ shows the signal in the time and frequency domain, respectively. The pressure signal collected from simulation or transducers is discrete. Thus, discrete time Fourier transform (DTFT) was used. Fast Fourier transform is usually used to compute DTFT efficiently in practical applications. In previous study, the frequency was used to determine the water hammer period and influence of boundary conditions, and the water hammer period was used to locate the downhole events.

The harmonic series of the water hammer fundamental frequency are odd in open boundary, which can be used to calculate the hydraulic length of the system according to Eq. (24),

$$L_2 = \frac{a(2n-1)}{4f_n}, \quad (24)$$

where L_2 is hydraulic length (m), n is the n^{th} harmonic of the fundamental frequency and f_n is the frequency of n^{th} harmonic (Hz). The hydraulic length shared the same value of the events' location from the wellhead (Clark et al. 2018).

2.2.3. Cepstrum analysis

Cepstrum analysis is used in nonlinear signal analysis and processing. This technique, first developed by Bogert in 1963, was used to

identify the echoes of seismic waves generated by earthquakes or bomb explosions (Parkhonyuk et al., 2018b). By taking the inverse Fourier transform of the logarithm of the Fourier transform of the signal, the cepstrum result is calculated according to

$$\hat{x}(\tau) = \frac{1}{2\pi} \int \log \left[\left| \int x(t) e^{i\omega t} dt \right| \right] e^{-i\omega\tau} d\omega, \quad (25)$$

where $x(t)$ and $\hat{x}(\tau)$ shows the signal in time and quefrequency domain, respectively.

In cepstrum, the independent variable τ is quefrequency and has the characteristics in the time dimension. In 2016, cepstrum analysis was first applied to analyze the water hammer signals during the fracturing operation, calculating the reflection time of downhole events. A time-frequency analysis method was employed to assess how the quefrequencies varied in time. The water hammer signal was broken into a series of windows of finite length and the cepstrum analysis was applied. Integrating the cepstrum results of all windows, the cepstrogram of the water hammer was calculated. The cepstrogram is a visual representation of the water hammer cepstrum matrix. Different downhole events correspond to the different polarity of cepstrum coefficients, and then the computed cepstrum matrix is divided into positive and negative cepstrum matrices. Choosing vectors having different physical time (t_i and t_j), the scalar product was taken to generate the following function,

$$F_{ij\tau} = V_{i,\tau} V_{j,\tau}. \quad (26)$$

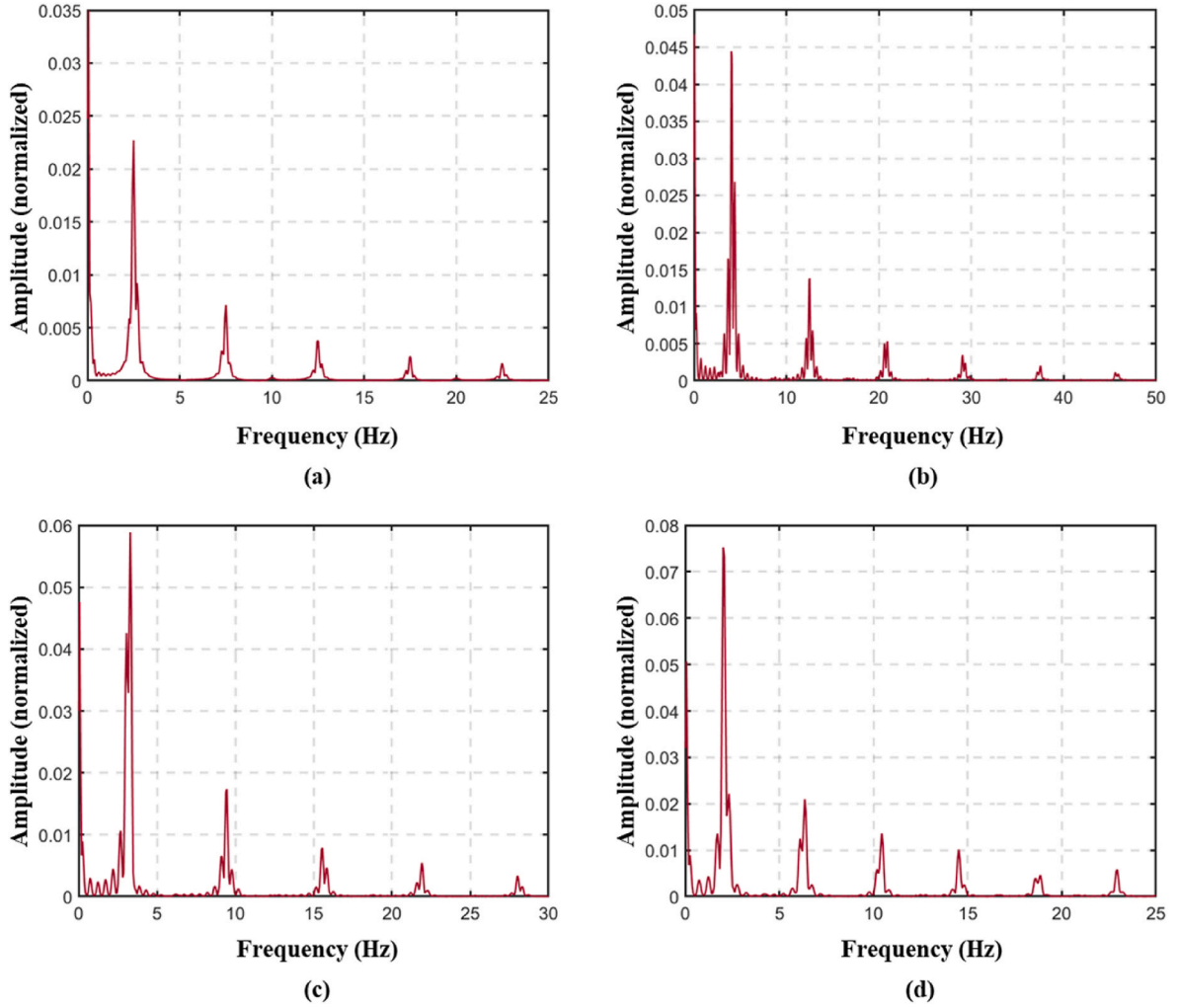


Fig. 10. The results of case 5 to 8 in the frequency domain: (a) case 5 fracture length: 200 m, location:100 m; (b) case 6 fracture length: 100 m, location:60 m; (c) case 7 fracture length: 100 m, location: 80 m; (d) case 8 fracture length: 100 m, location:120 m.

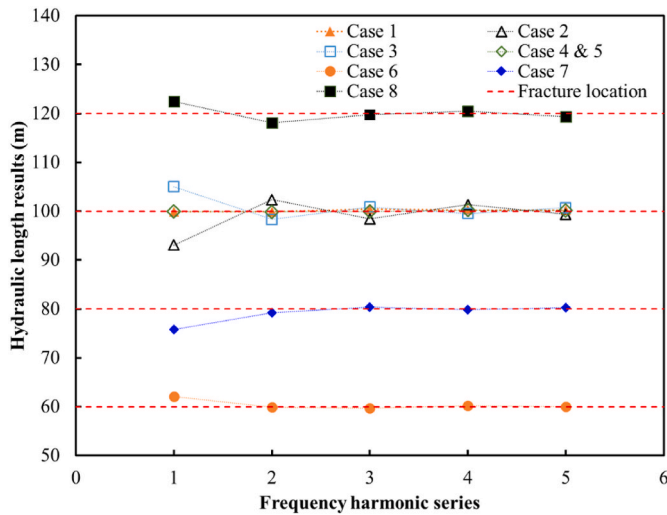


Fig. 11. The fracture location estimation calculated by frequency harmonic series.

Here i, j correspond to two different points at different physical times, and τ means a centroid of each vector in the reflection time domain. Taking the sum of all vectors with the different physical time (t), the function F_τ of reflection time would be generated (Korkin et al. 2020),

$$F_\tau = \sum_{ij} F_{ij} = \sum_{ij} V_i V_j. \quad (27)$$

The quefrequency τ of the peaks in F_τ represents the reflection time of downhole events. Fig. 2 below shows the process of the functions of reflection time F_τ calculation.

The positive and negative cepstrum coefficient matrices were then calculated to obtain F_τ^+ and F_τ^- . In previous study, the reflections of hydraulic-stimulated fractures (negative reflection events) are manifested by strong negative peaks in F_τ^- . Similarly, the reflections of wellbore restriction (positive reflection event) result in strong positive peaks in F_τ^+ (Bogdan et al., 2016; Parkhonvuk et al., 2018b). The location of the reflection events L_3 can be calculated by (Parkhonvuk et al. 2018b):

$$L_3 = \frac{a \times \tau}{2}. \quad (28)$$

2.3. Model configuration

During the hydraulic fracturing treatment, fluid is injected into the

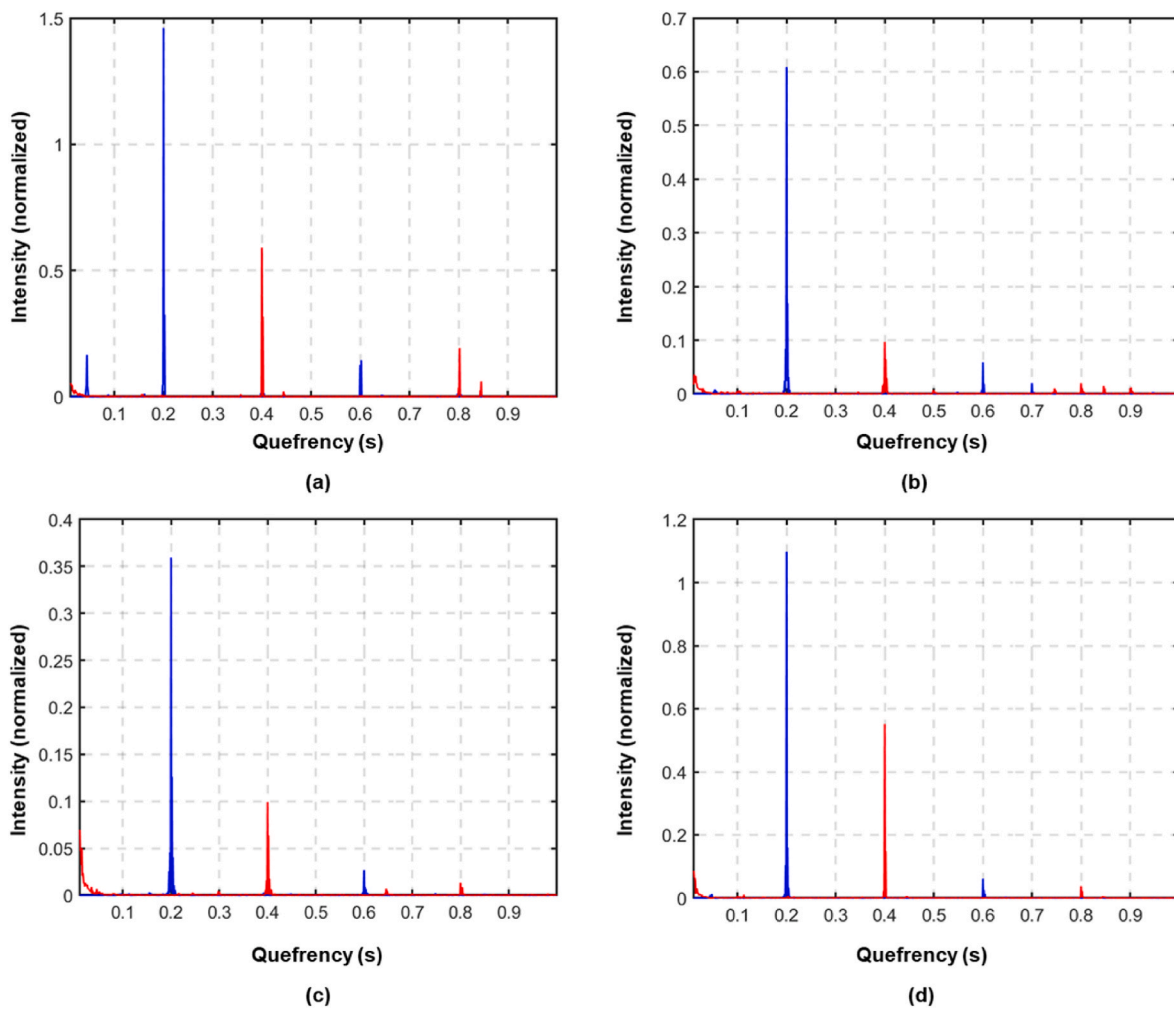


Fig. 12. The results of case 1 to 4 in the quefrequency domain: (a) case 1 fracture length: 0 m, location: 100 m; (b) case 2 fracture length: 25 m, location: 100 m; (c) case 3 fracture length: 50 m, location: 100 m; (d) case 4 fracture length: 100 m, location: 100 m.

wellbore by fracturing pumps at the surface. Fluid flows in the wellbore first and then through the perforation holes into the formation. After the pump shuts down, the pressure fluctuation is generated in the wellbore called the water hammer. Pressure wave propagates rapidly in the wellbore and reflects after interacting with downhole events. In this model, we designed a linear discharge decrease as the inlet boundary. The pump shut-in time was short to get a direct water hammer. At the other end of the wellbore, a zero-flow boundary condition was assigned, representing the well bottom, the bridge, or the plug. To consider the influence of that pressure wave propagating in the fracture, we assumed the hydraulic connected fracture of the wellbore as an equivalent branch pipe. A pressure boundary was set at the outlet of the branch pipe to simulate the fracture connection to the formation. These boundaries enable the fluid flow in the pipeline system to match the actual conditions. The fracture location was defined as the distance between the fracture and the wellhead, and the fracture length was the length of the branch pipe. As shown in Fig. 3, the whole numerical configuration included a wellbore and a branch-pipe-like fracture.

The output result of the model calculation was the water hammer pressure fluctuation at the wellhead in the wellbore-fracture system within 10 s. After normalizing the pressure, a multi-dimensional analysis method was used to study the response characteristics of the water hammer wave in the wellbore-fracture system.

According to the previous research, the pressure wave speed is varied in the wellbore and the fracture (Carey et al., 2015; Liang et al., 2017; Dung et al., 2021). The pressure wave propagating in the Fluid-filled

fractures was known as Krauklis waves, which was much slower than the tube wave. In this work, the wave speed in the wellbore and the hydraulic-connected fracture were assumed as 1000 m/s and 100 m/s, respectively.

2.4. Model validation

Before applying this transient flow model to study the response characteristics of the water hammer in a wellbore-fracture system, the computational fluid dynamics (CFD) software FLUENT was used to validate this model. FLUENT is a widely used CFD software at present and has good reliability. The basic parameters in model validation are shown in Table 1.

When the inlet discharge decreased, the pressure in the system began to drop rapidly, and the water hammer wave was generated at the inlet and propagated along the pipe. Fig. 4 shows the simulation results comparison of the FLUENT and our model solved by MOC in this study. By comparing the normalized pressure data, there was a good agreement in water hammer signatures, including the shape, period, and amplitude. The cross-correlation coefficient of the two results was 99.8%, and the mean square error was 0.055%. The FLUENT simulation result is consistent with the results of our model. The MOC method can calculate the pressure fluctuation in a very short time rather than the CFD method. Therefore, the model presented in this paper is effective in simulating the propagation of the water hammer in the wellbore-fracture system.

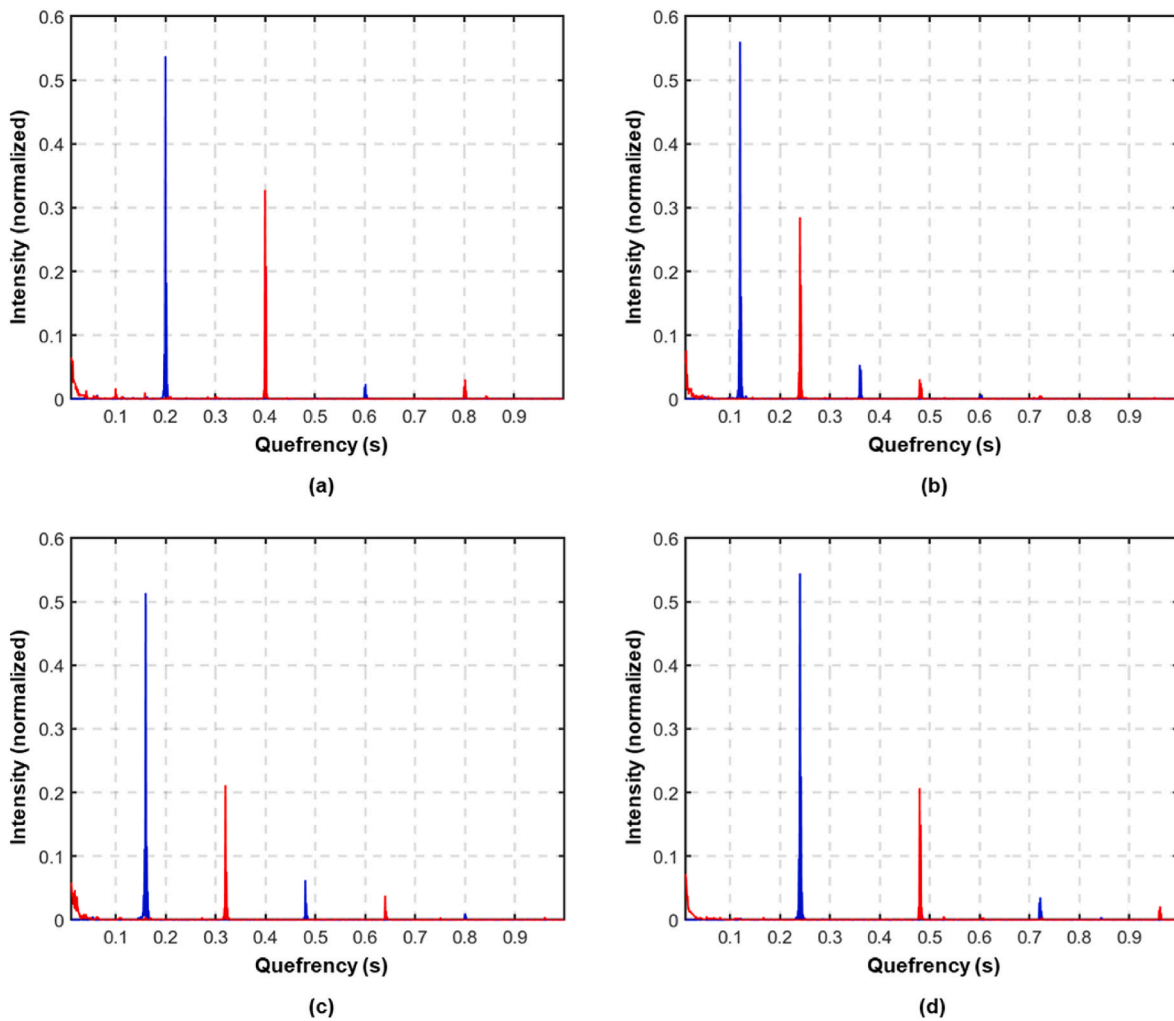


Fig. 13. The results of case 5 to 8 in the queffrequency domain: (a) case 5 fracture length: 200 m, location:100 m; (b) case 6 fracture length: 100 m, location:60 m; (c) case 7 fracture length: 100 m, location: 80 m; (d) case 8 fracture length: 100 m, location:120 m.

3. Results and discussion

In this section, we presented eight cases to study the water hammer response in the wellbore-fracture system in time, frequency and queffrequency domain respectively, while the existence of the fracture, the location and the length of the stimulated fracture were varied. The parameters and boundary conditions used in the simulation are shown in Table 2. Case 1 and 4 compared the effect of the fracture existence on water hammer response. Case 2 to 5 compared the effect of varied fracture length on water hammer response. Case 4 and case 6 to 8 compared the effect of varied fracture location on water hammer response.

In case 1, with zero-meter fracture, the outlet consistent pressure boundary is directly located at the position of the fracture. The wellbore-fracture system can be equivalent to a wellbore with a length of 100 m, as shown in Fig. 5.

3.1. Water hammer response in the time domain

The time-domain response of the water hammer wave without the influence of the hydraulic-connected fracture is shown in Fig. 6(a). As the water hammer pressure wave propagated, the water hammer response gradually attenuated. When the system was converted into the wellbore-fracture system, the response of the water hammer in the time domain changed significantly, as shown in Figs. 6 and 7. The pressure

signatures had more obvious attenuation and were affected with other pressure pulses. The longer the fracture was, the more obvious the pressure attenuation would be.

As the discharge decreased, the pressure dropped instantly and sharply. When the value was completely reduced to zero, the trend of the pressure drop changed. The results of all cases simulated in this work showed the same that the time of the first pressure response was 0.01 s. It was consistent with the time the discharge value was set from the initial condition reducing to zero. Therefore, the first water hammer pressure response in the time domain was affected by the pump shut-in time in the direct water hammer of this work. The change of pressure during the pump shut-in time represents the cumulative pressure change at this time. Thus, the water hammer pressure signatures will be influenced by the shut-in time (the setting in this work was 0.01 s) under the direct water hammer condition.

In case 1 to 5 (fracture with the same location but different length), the second pressure change-point were consistent with 0.2 s. In case 6 to 8 (fracture with the same length but different location), the time of the second change-point was 0.12, 0.16, and 0.24 s. The reflection position of all eight cases, calculated from Eq. (22), shared the same value with the fracture location. Therefore, the second pressure response time of the water hammer was determined by the position of the fracture. The third pressure response time, 0.01 s after the second one, was the effect of the shut-in time.

The water hammer pressure trend in case 1 was the same in the

following moment, changing only at every 0.2 s (boundary reflection time) and 0.01 s (pump shut-in time). When the system condition converted into the wellbore-fracture system in case 2 to 8, the time-domain response of water hammer pressure was different from that of case 1 because of the addition of the fracture and well bottom. In case 2 to 8, the fourth pressure change moment was 0.3 s, which matched the time of the well bottom reflection.

The pressure wave propagates in the fracture as a slow wave. Therefore, the effect of the fracture on the pressure wave response was much slower than that of the tube wave in the wellbore. When the water hammer wave propagated in the fracture and reflects, taking case 4 for instance, the attenuation of the water hammer pressure changes at the time of 2.2 s. The trend of water hammer pressure attenuation changed every 2 s in case 2 and case 6 to 8 (fracture with the same length), which was consistent with the time of water hammer wave reflected in the fracture. Other cases had the same changing law of this pressure pulse affected by the fracture, which responded every 0.5 s, 1 s, and 4 s in case 2, case 3, and case 5, respectively. The pressure pulse from the fracture was labeled in Figs. 6 and 7 with blue arrows.

In case 8, significant oscillations between the fracture and the bridge plug were discovered. The duration of the first and the second pressure reflection time between the fracture and the bridge plug were studied. Between 0.24 s and 0.48 s, the pressure fluctuation occurred every 0.06 s, excluding the influence of pump shutdown time of 0.01 s. The calculated reflection distance was consistent with the distance between the well bottom and the fracture. This pressure fluctuation was obvious before the oscillation of the fracture traveled back to the wellhead. When the distance between the fracture to the well bottom and the distance of the fracture to the wellhead was similar (80 m and 70 m in case 7, respectively), these two oscillations would affect each other. The pressure response time marked in case 7 was 0.02 s, which was the difference between the reflection time of these two oscillations. These two discoveries were labeled in Fig. 8 with blue circles.

3.2. Water hammer response in the frequency domain

In the frequency domain, the amplitude peaks were considered as the frequency of the water hammer response. Under the condition of the wellbore and wellbore-fracture system, the frequency responses of the water hammer were similar and multiple repeating peaks were shown. The energy of the fundamental frequency was the strongest when the fracture length was zero. The harmonic series of the water hammer resonance was shown in Fig. 9 and Fig. 10. All the spectrums had a very strong zero-frequency component. The frequency series of the water hammer signal occurred at the odd-integer multiples of the fundamental frequency, which was consistent with previous studies. The higher-frequency components represented the high-order harmonic. In addition, when the fracture was introduced to the system, amplitude peaks at different frequencies appeared in the spectrum of the water hammer signal, located beside the main events' peaks. The reason was that the addition of the fracture brings different types of vibration into the system, which led to the modulation of the water hammer signals.

Fig. 11 presents the hydraulic length calculated by the fundamental frequency and its' harmonic series of the water hammer fluctuation. Case 1 (no fracture) and case 4 (fracture with 100 m length) have the same fundamental and harmonic series. According to Eq. (25), the hydraulic length results were calculated and the fracture location was estimated. The fracture locations of all cases were almost the same, whereas errors were acceptable. In case 2 (fracture with 25 m length) and 3 (fracture with 50 m length), due to the shorter fracture, the water hammer reflection within the fracture returned rapidly to the wellbore and affected the water hammer response. Although the inlet position was the same as case 1 and case 2, the fundamental frequency and harmonic sequence were different, the calculated hydraulic length and errors of the fracture location were larger than those in case 1 and case 4. The calculation errors of the fundamental frequency were the largest,

and the errors decreased with the increase of the harmonic series.

In case 4 and 5 (fracture with 100 m and 200 m length), when the fracture length was long, the internal reflection of the fracture had little influence on the wellbore water hammer response. The fundamental frequency and harmonic sequence were the same as those in case 1, and the calculated hydraulic length and fracture location were also the same. In case 4 and case 6 to 8 (fracture with different location), the values of fundamental frequency and harmonic series became smaller as the fracture location became further from the wellhead. The hydraulic length was consistent with the fracture location under acceptable errors. The response of oscillations between the fracture and the well bottom was not obvious in the spectral results.

The response characteristics of the water hammer signal in the frequency domain represented the flow entrance in the wellbore-fracture system. The calculated hydraulic length represented the fracture location. The calculation errors of hydraulic length by harmonic frequency were smaller than those by the fundamental frequency. The possible reason was that the harmonics were slightly affected by the water hammer attenuation than the fundamental frequency.

3.3. Water hammer response in the quefrequency domain

In the cepstrum results, the response of the fracture appeared as non-zero peaks at T , $2T$, $3T$..., where T represents the reflection time, and the polarity changes (Parkhonvuk et al., 2018b). According to the results of water hammer reflection functions in this paper, which were shown in Fig. 12 and Fig. 13, the first peak was the negative response with the maximum intensity. At the moment doubling the time of the first peak appeared positive response, which was consistent with the results of former research.

In case 1 to 5 (fracture with different length but same location), the moment corresponding to the negative response was 0.2 s, consistent with the time of the second pressure change point in the time domain. The reflection position was found to be 100 m from the inlet by Eq. (28), which was consistent with the fracture location. The moment corresponding to the positive response was 0.4 s, which was twice the time of the negative response. The third and the fourth response moment also followed this rule. With the change of fracture length, the response time of the cepstrum remained unchanged, which represented the reflection response of the fracture mouth. In case 4 and case 6 to 8 (fracture with different location), the response time changed with the increase of the fracture location, but the reflection position corresponding to the reflection time was always the location of the fracture mouth. In the cepstrum results, the oscillations between the fracture and the bridge had no obvious characteristics. Regardless of the change of the fracture length or the fracture location, the response characteristics of the water hammer in the quefrequency domain represented the flow entrance in the wellbore-fracture system, which was the length from the fracture mouth to the wellhead.

4. Conclusions

In this work, we proposed a multi-dimensional analysis method in time, frequency and quefrequency domain for water hammer pressure analysis. Meanwhile, the water hammer pressure response characteristics of the wellbore-fracture system were sufficiently presented for the first time to accurately investigate the water hammer propagation in the wellbore-fracture system. The model was built on the transient flow model and the fracture was added in as a branch pipe to the wellbore, then verified by FLUENT. Eight cases were designed to analyze and summarize the water hammer response characteristics in the wellbore-fracture system, where the fracture existence, location, and length of the fractures were varied. The conclusions can be summarized as follows:

1. In the time domain, the existence of the fracture will enhance the attenuation of the water hammer wave. When the shut-in time is short and a direct water hammer is generated, the reflections of the water hammer interacting with the fracture mouth, well bottom and the fracture toe have a clear manifestation, which alters the pressure trend.
2. In the time domain, the oscillation between the fracture mouth and the bridge has an obvious response in the water hammer signal. Moreover, this pressure fluctuation behavior is obvious before the pressure oscillation from the inlet to the fracture mouth is back to the wellhead. If the length of the inlet to the fracture mouth and the length of the fracture mouth to the well bottom are similar, these two oscillations will affect each other and additional pressure oscillation will appear in the time domain.
3. In the frequency domain, the peaks in the spectrum of the water hammer signal represent the flow entrance in the wellbore-fracture system. The fracture introduced to the system will lead to spectrum energy reduction. The calculated hydraulic length represents the fracture location. The calculation errors of hydraulic length calculated by harmonic frequency are smaller than those by the fundamental frequency.
4. In the quefrency domain, the response characteristics of the water hammer represent the flow entrance in the wellbore-fracture system, which is consistent with the fracture location.
5. The results of the multi-dimensional analysis are closely related to the flow entrance of the fracture, which is hydraulically connected to the wellbore. They should be considered together to accurately locate the fracture position and investigate more downhole events when water hammer fracture diagnostics is applied.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to acknowledge the financial support from the National Natural Science Foundation of China (No. 52004300).

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